Azure Fundamentals Lab Manual

Hands-on Lab: Virtual Machine, Azure SQL Database, Blob Storage, and a Simple HTML Website

All exercises use the Azure Portal (no command line required).

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Introduction

This lab manual is a guided, hands-on introduction to four core Azure services. Every step is performed in the Azure Portal using point-and-click navigation — no scripting or command-line tools are required. By the end of this lab you will have created a virtual machine, an Azure SQL database, a storage account with image upload, and a simple HTML website hosted on Azure.

What You Will Build

Lab	Outcome
Lab 1: Virtual Machine	A running Windows or Linux VM you can connect to remotely
Lab 2: Azure SQL Database	A cloud SQL database with a server, accessible via Query editor
Lab 3: Blob Storage	A storage account and container with an image file uploaded
Lab 4: Simple HTML Deployment	A static HTML page published to a public URL

Prerequisites

- An active Azure subscription (a free trial or pay-as-you-go subscription both work).
- Permission to create resources in at least one resource group.
- A modern web browser (Edge, Chrome, or Firefox).
- One small image file (JPG or PNG) saved on your computer, for Lab 3.
- A short HTML file, for Lab 4 (a sample is provided in Lab 4).

Warning — Cost Control
Every resource created in this lab manual incurs cost while it is running. Follow the Cleanup Checklist at the end of this document to delete resources when you are finished, especially the virtual machine and the SQL database.

Conventions Used in This Manual

- Numbered steps describe actions to take in the Azure Portal.
- Yellow Note boxes provide helpful context or alternatives.
- Red Warning boxes flag cost or security considerations.
- Text in quotation marks (e.g. “Create a resource”) refers to exact buttons, menu items, or labels in the portal.

Lab 1: Create a Virtual Machine

In this lab, you will deploy a virtual machine (VM) running Windows Server or Ubuntu Linux directly from the Azure Portal.

Step 1: Sign In and Start the Wizard

1. Go to the Azure Portal at portal.azure.com and sign in with your Azure account.
2. In the search bar at the top of the page, type “Virtual machines” and select it from the results.
3. On the Virtual Machines page, select “+ Create”, then choose “Azure virtual machine” from the dropdown.

Step 2: Configure the Basics Tab

4. Under Project details, select your Subscription.
5. For Resource group, select “Create new” and enter a name, for example LabResourceGroup.
6. Under Instance details, enter a Virtual machine name, for example MyLabVM.
7. Choose a Region close to you.
8. For Image, select a Windows Server image (e.g. Windows Server 2022 Datacenter) or Ubuntu Server (e.g. Ubuntu Server 22.04 LTS).
9. Leave the default Size, or select “See all sizes” and choose a small, low-cost size such as B1s for lab purposes.
10. Under Administrator account, enter a Username and a Password (or upload an SSH public key if using Linux). The password must be at least 12 characters and meet complexity requirements.
11. Under Inbound port rules, allow the relevant inbound port: RDP (3389) for Windows, or SSH (22) for Linux.

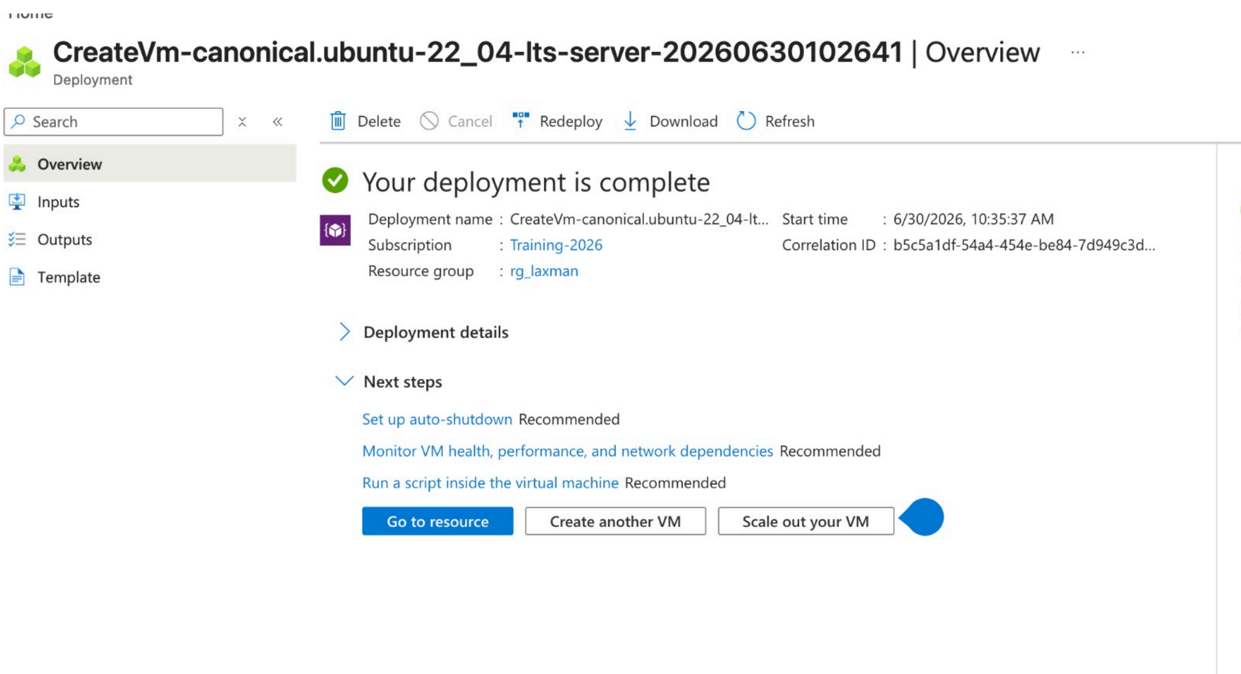
Step 3: Review and Create

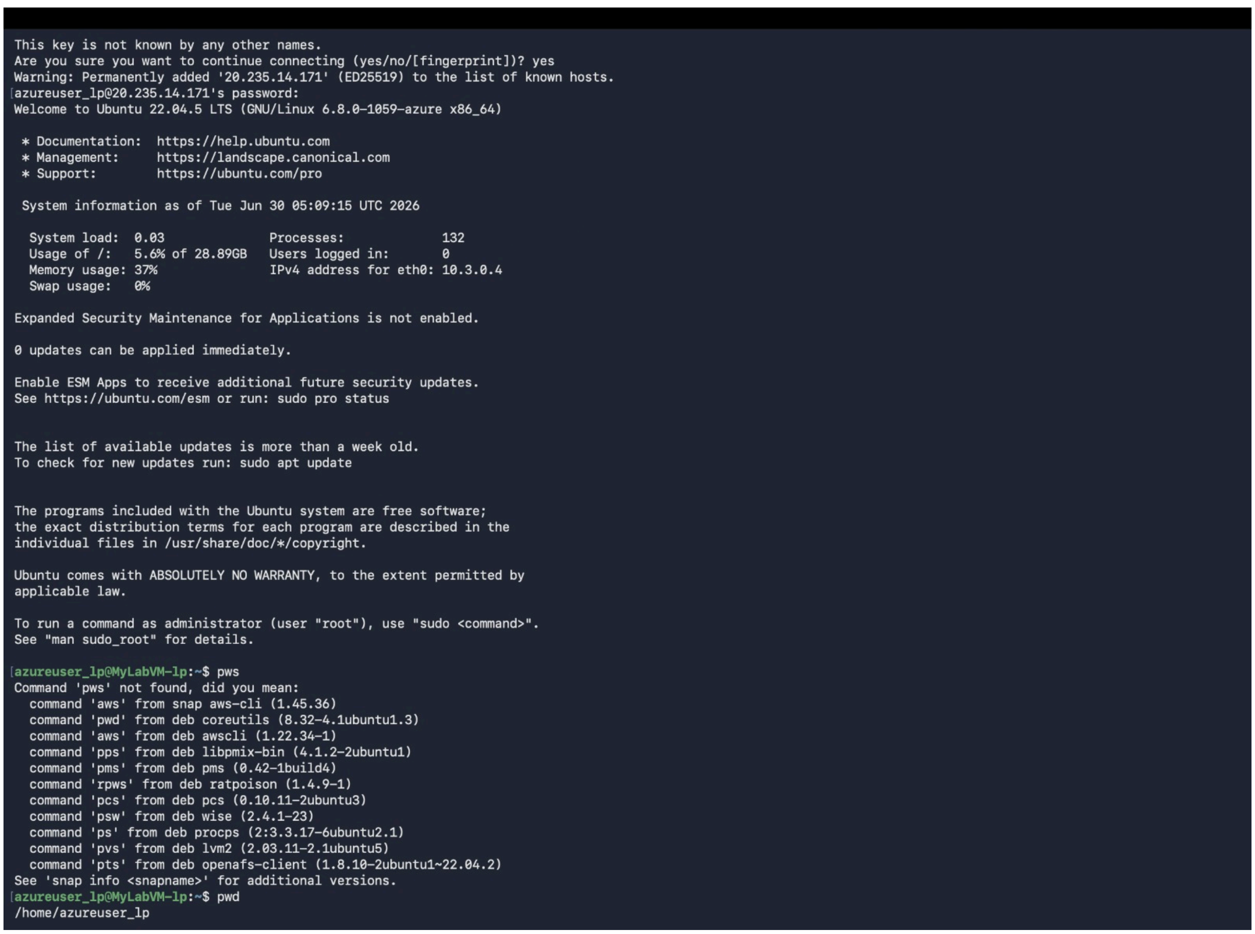
12. Select “Review + create” at the bottom of the page.
13. Wait for validation to pass, then review the configuration summary.
14. Select “Create”.
15. Wait for the deployment to complete (typically 1–3 minutes), then select “Go to resource”.

Step 4: Connect to the VM

16. On the VM Overview page, select “Connect”.
17. For Windows, choose RDP, download the RDP file, and open it with Remote Desktop Connection using the username and password you set.
18. For Linux, choose SSH and follow the connection command shown, using an SSH client such as PuTTY or a terminal.

Note
If RDP or SSH connection fails, confirm the inbound port rule was created and that your network allows outbound connections on that port.





```
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.235.14.171' (ED25519) to the list of known hosts.
azureuser_lp@20.235.14.171's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1059-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Jun 30 05:09:15 UTC 2026

  System load:  0.03              Processes:             132
  Usage of /:   5.6% of 28.89GB   Users logged in:       0
  Memory usage: 37%               IPv4 address for eth0: 10.3.0.4
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser_lp@MyLabVM-lp:~$ pws
Command 'pws' not found, did you mean:
  command 'aws' from snap aws-cli (1.45.36)
  command 'pwd' from deb coreutils (8.32-4.1ubuntu1.3)
  command 'aws' from deb awscli (1.22.34-1)
  command 'pps' from deb libpmix-bin (4.1.2-2ubuntu1)
  command 'pms' from deb pms (0.42-1build4)
  command 'rpws' from deb ratpoison (1.4.9-1)
  command 'pcs' from deb pcs (0.10.11-2ubuntu3)
  command 'psw' from deb wise (2.4.1-23)
  command 'ps' from deb procps (2:3.3.17-6ubuntu2.1)
  command 'pvs' from deb lvm2 (2.03.11-2.1ubuntu5)
  command 'pts' from deb openafs-client (1.8.10-2ubuntu1~22.04.2)
See 'snap info <snapname>' for additional versions.
azureuser_lp@MyLabVM-lp:~$ pwd
/home/azureuser_lp
```

Lab 2: Create an Azure SQL Database

In this lab, you will create a logical SQL server and a single Azure SQL Database, then query it using the built-in Query editor — entirely in the browser.

Step 1: Start the Wizard

19. In the Azure Portal search bar, type “SQL databases” and select it.
20. Select “+ Create”.

Step 2: Configure the Basics Tab

21. Select your Subscription.
22. For Resource group, select the same resource group used in Lab 1, or create a new one.
23. Under Database details, enter a Database name, for example mySampleDatabase.
24. Next to Server, select “Create new”. Enter a unique Server name, choose a Location, select an authentication method (SQL authentication is simplest for a lab), and set a Server admin login and Password. Select “OK”.
25. Leave “Want to use SQL elastic pool?” set to No.
26. Under Compute + storage, select “Configure database”, choose the General Purpose tier with Serverless compute, and select the smallest vCore option to minimize cost. Select “Apply”.
27. Under Backup storage redundancy, choose Locally-redundant backup storage (lowest cost for a lab).

above 3 points not needed if chosen free offer

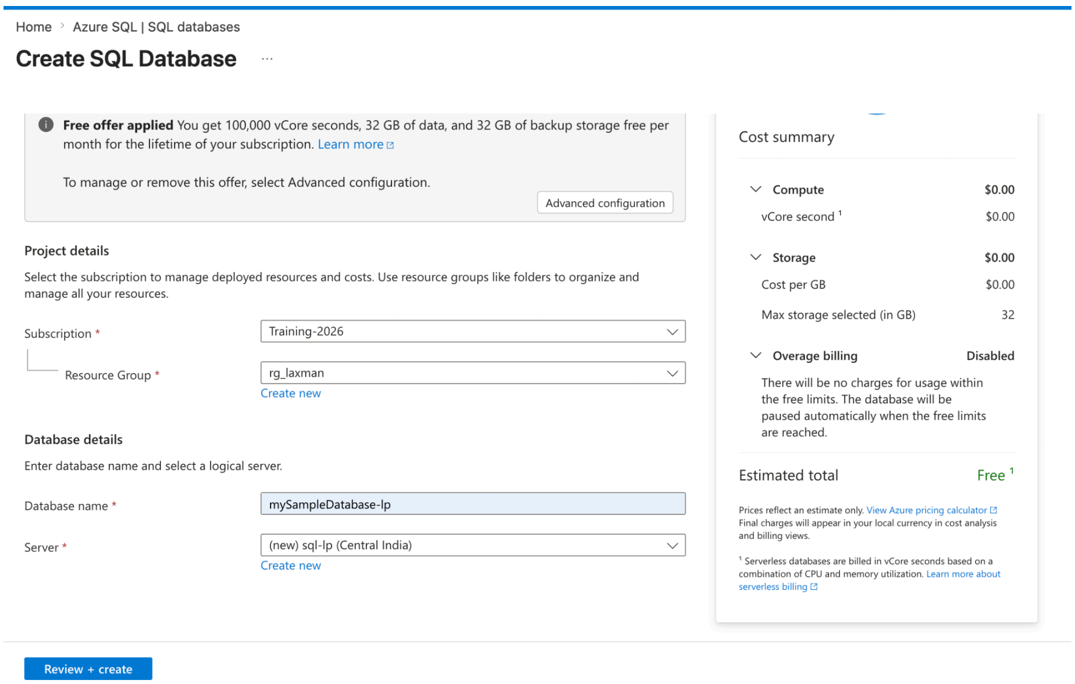
Step 3: Configure Networking

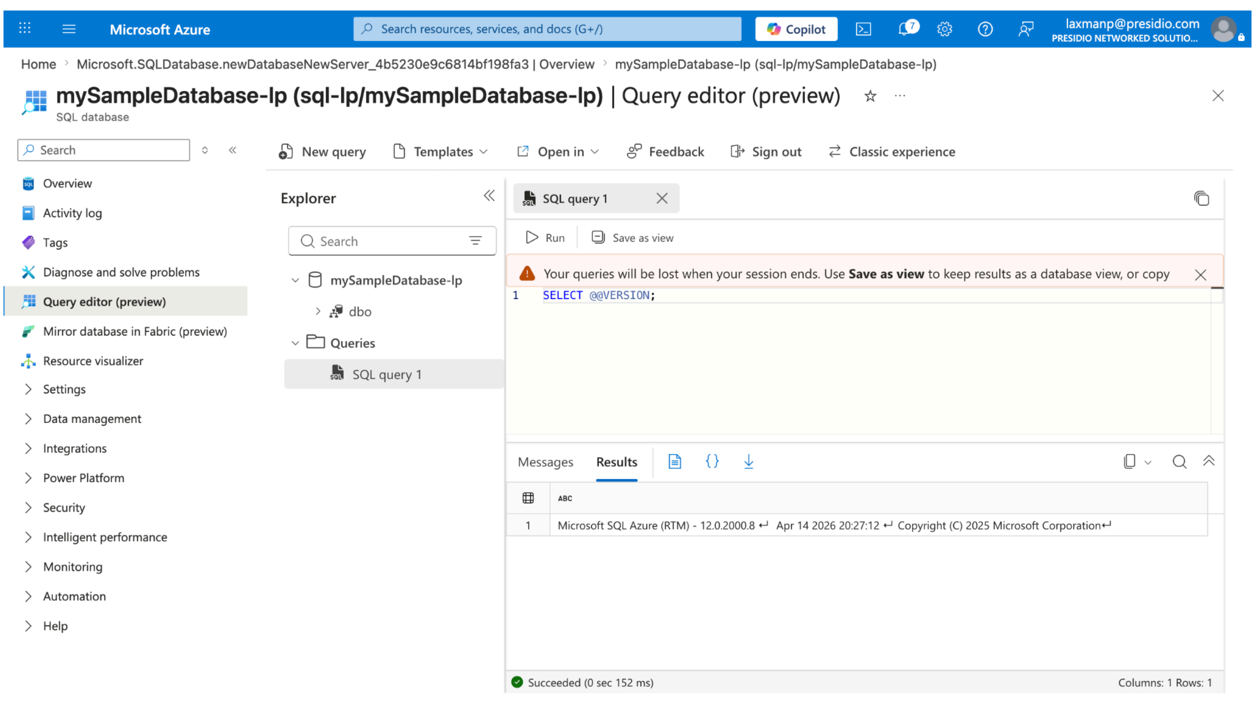
28. Select the “Networking” tab.
29. For Connectivity method, select “Public endpoint”.
30. Set “Allow Azure services and resources to access this server” to Yes.
31. Set “Add current client IP address” to Yes so you can connect from your own machine.

Step 4: Review, Create, and Query

32. Select “Review + create”, wait for validation, then select “Create”.
33. Once deployment finishes, select “Go to resource”.
34. On the database Overview page, select “Query editor (preview)” in the left menu.
35. Sign in using the SQL admin login and password you created.
36. In the query window, run a simple test query, for example: SELECT @@VERSION;, then select “Run” to confirm the database is working.

Note
The Query editor runs entirely in the browser, so no database client software is required for this lab.





Lab 3: Blob Storage and Image Upload

In this lab, you will create a storage account, add a container, and upload an image file as a blob.

Step 1: Create a Storage Account

37. In the Azure Portal search bar, type “Storage accounts” and select it.
38. Select “+ Create”.
39. Select your Subscription and the resource group from Lab 1.
40. Enter a globally unique Storage account name (lowercase letters and numbers only, e.g. mylabstorage2026).
41. Choose a Region close to you.
42. Leave Performance set to Standard and Redundancy set to Locally-redundant storage (LRS) for a low-cost lab setup.
43. Select “Review”, then “Create”, and wait for deployment to finish.
44. Select “Go to resource”.

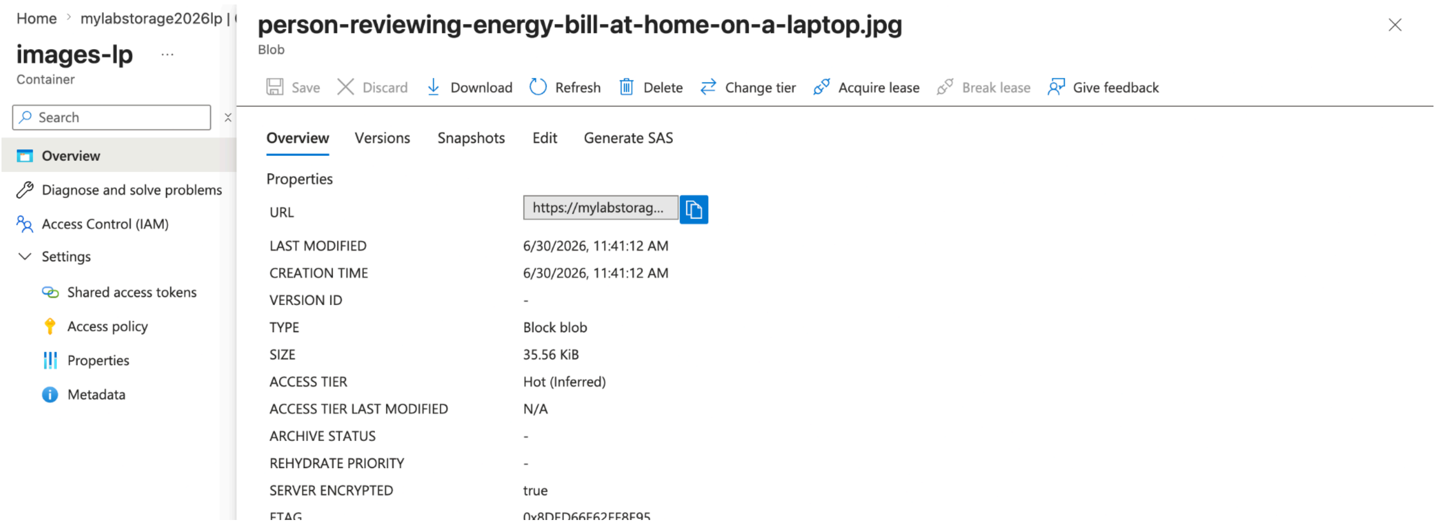
Step 2: Create a Container

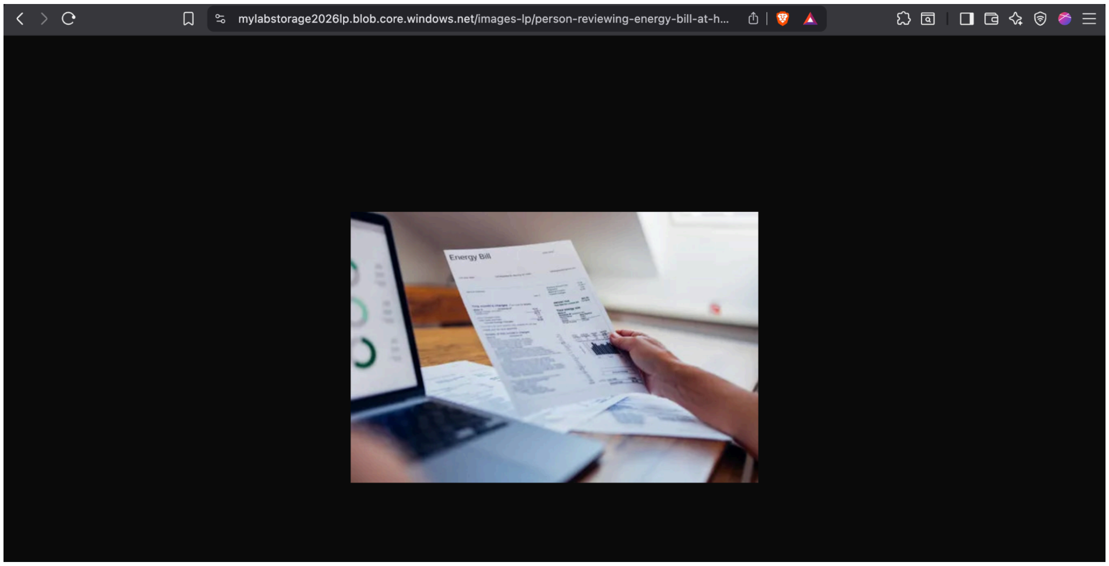
45. In the left menu of the storage account, under Data storage, select “Containers”.
46. Select “+ Container”.
47. Enter a name, for example images (lowercase letters, numbers, and dashes only).
48. Leave the default Public access level (Private), then select “Create”.

Step 3: Upload an Image

49. Select the new container to open it.
50. Select “Upload”.
51. Browse to the image file on your computer (JPG or PNG) and select it.
52. Select “Upload” to complete the operation.
53. Select the uploaded blob in the list to view its properties and copy its URL.

Note
If the container's public access level is Private, the blob URL will not be publicly viewable in a browser. To make an individual image viewable via its URL, set the container access level to “Blob” (anonymous read access for blobs only) when creating it, or change it afterward under container Settings > Access level.


Home
mylabstorage2026lp |
images-lp
Container
Search
Overview
Diagnose and solve problems
Access Control (IAM)
Settings
Shared access tokens
Access policy
Properties
Metadata
person-reviewing-energy-bill-at-home-on-a-laptop.jpg
Blob
Save
Discard
Download
Refresh
Delete
Change tier
Acquire lease
Break lease
Give feedback
Overview
Versions
Snapshots
Edit
Generate SAS
Properties
URL
https://mylabstorag...
LAST MODIFIED
6/30/2026, 11:41:12 AM
CREATION TIME
6/30/2026, 11:41:12 AM
VERSION ID
-
TYPE
Block blob
SIZE
35.56 KiB
ACCESS TIER
Hot (Inferred)
ACCESS TIER LAST MODIFIED
N/A
ARCHIVE STATUS
-
REHYDRATE PRIORITY
-
SERVER ENCRYPTED
true



Lab 4: Deploy a Simple HTML Website

In this lab, you will publish a basic HTML page to the public internet using the static website feature of the storage account you created in Lab 3 — all from the portal, with no command-line deployment tools.

Step 1: Prepare a Sample HTML File

Create a text file named index.html on your computer with the following content, then save it:

```
<!DOCTYPE html> <html> <head><title>My Azure Lab Site</title></head> <body>   <h1>Hello from
Azure Blob Storage!</h1>   <p>This page was deployed using static website hosting.</p> </body>
</html>
```

Step 2: Enable Static Website Hosting

54. Open the storage account created in Lab 3.
55. In the left menu, under Data management, select “Static website”.
56. Select “Enabled”.
57. For Index document name, enter index.html.
58. Select “Save”.
59. Azure displays a Primary endpoint URL once saved — copy this URL; it is the public address of your site.

Step 3: Upload the HTML File

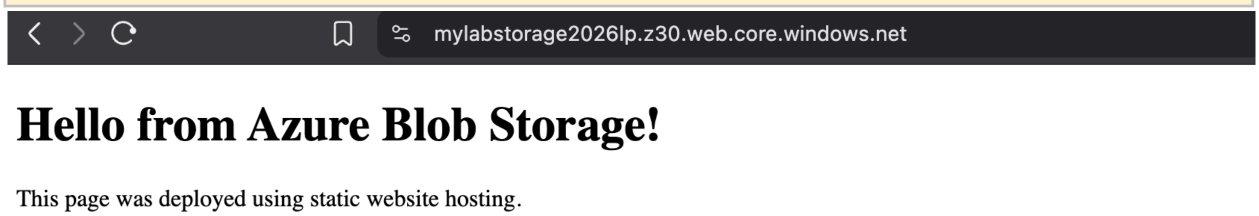
60. After saving, return to the Static website page and select the link next to “$web container” (Azure automatically creates a container named $web for static content).
61. Select “Upload”.
62. Browse to your index.html file and select it.

63. Select “Upload” to complete the operation.

Step 4: View the Site

64. Return to the Static website page in the left menu of the storage account.
65. Select or copy the Primary endpoint URL and open it in a new browser tab.
66. Confirm that your HTML page renders correctly.

> **Note**
> The $web container is automatically configured for public, anonymous read access when static website hosting is enabled, so your page is reachable by anyone with the URL.



Cleanup Checklist

Azure resources continue to incur charges until they are deleted. When you have finished the lab, clean up resources to avoid unnecessary cost.

Option A: Delete the Entire Resource Group (Recommended)

67. In the Azure Portal search bar, type “Resource groups” and select it.
68. Select the resource group used throughout this lab (e.g. LabResourceGroup).
69. Select “Delete resource group”.
70. Type the resource group name to confirm, then select “Delete”.

Deleting the resource group removes the VM, SQL server and database, storage account, and all associated resources (disks, network interfaces, public IP addresses) in one step.

Option B: Delete Resources Individually

- Virtual machine: open the VM, select “Delete”, and also delete the associated disk, network interface, public IP, and network security group shown on the VM's Overview page.
- SQL Database and server: open the SQL server resource, select “Delete” (this also offers to delete databases on that server).
- Storage account: open the storage account, select “Delete”, and confirm by typing the account name.

> **Warning — Cost Control**
> A virtual machine continues to bill for compute time while running, and for attached disks even when stopped. Always delete or fully deallocate resources you no longer need.

Summary

In this lab manual you created and configured four foundational Azure resources entirely through the Azure Portal: a virtual machine, an Azure SQL Database, a Blob Storage container with an uploaded image, and a static HTML website. These exercises cover compute, database, and storage — three of the core pillars of the Azure platform — and provide a foundation for more advanced labs involving networking, security, and automation.

Further Reading

Create a Windows VM in the Azure portal

Create a single Azure SQL database (portal)

Upload, download, and list blobs (Azure portal)

Host a static website in Azure Storage